Nutrition Assistant

Final Project Demonstration Report

Date	18 July 2026
Team ID	
Project Name	Nutrition Assistant
Team Leader	Naraharisetti Sai Nitchal
Team Members	Sundarapalli Siva Sankar, Edward Jones Devathoti, Nunna Tarun Sri Venkata Sai, SAIRAJU ADDAGARLA
Technology Stack	MongoDB · Express.js · React.js (Vite) · Node.js (MERN)

1. Introduction

1.1 Project Overview

Nutrition Assistant is a full-stack MERN (MongoDB, Express.js, React.js, Node.js) web application for comprehensive lifestyle, diet and water tracking. It gives users exact physical calculators, a structured daily tracker, recipe discovery, meal planning, and an AI Nutrition Chatbot — all tied to one personal profile, covering the complete journey of understanding one's body, tracking intake, and getting personalized guidance on a single platform.

The application is built as a three-tier system: a React 19 + Vite client for the presentation layer, an Express.js/Node.js REST API for business logic, and MongoDB (via Mongoose) for persistent data storage. The client is deployed as a static site while the server runs as a persistent Node.js service connected to a managed MongoDB Atlas cluster.

1.2 Purpose

Nutrition Assistant aims to replace the fragmented mix of calculators, food diaries and generic online diet advice with one centralized, accurate and personalized platform. Traditional approaches to nutrition tracking suffer from disconnected tools, manual calculation, and generic (non-personalized) guidance — Nutrition Assistant addresses this gap with an ntegrated experience built around a single user profile.

2. Ideation Phase

2.1 Problem Statement

PS	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	a health-conscious individual trying to manage my diet	track my daily calories, macros and water intake accurately	I can't tell how many calories or how much water I actually need	nutrition calculators, food logging and expert guidance all live in separate, disconnected tools	overwhelmed and likely to give up on my health goals
PS-2	someone trying to lose/gain weight or manage a diet preference	get a meal plan and nutrition advice tailored to my body and goals	generic diet advice online doesn't account for my personal profile	most free tools don't personalize recommendations or answer questions instantly	frustrated and unsure whether I'm making the right food choices

2.2 Empathy Map Canvas

Persona: The Health-Conscious User

Dimension	Insights
Think & Feel	Anxious about whether their diet is balanced; wants a simple way to know if they are on track without a personal nutritionist.
See	Conflicting diet advice online; calorie apps cluttered with ads and paywalls.
Hear	Friends' contradictory BMI/TDEE numbers and fad-diet stories; trainers pushing supplements.
Say & Do	Repeatedly searches "how many calories should I eat"; logs food manually in a notes app; forgets to track water.
Pain	No centralized tracker tying body stats, food log and water log together; generic advice ignores personal goals.
Gain	A single dashboard with accurate calculators, effortless logging, and instant personalized guidance.

2.3 Brainstorming

Ideas generated during ideation, grouped and prioritized by importance and feasibility:

- BMI & BMR/TDEE calculators with instant, real-time results (High importance, High feasibility)
- Daily food & water tracker with auto-computed totals (High importance, High feasibility)
- Dashboard with weekly trends & goal-based recommendations (High importance, Medium feasibility)
- Recipe explorer with search, filters & favorites (Medium importance, High feasibility)
- Meal planner for scheduling meals by time of day (Medium importance, Medium feasibility)
- AI Nutrition Chatbot for personalized diet Q&A (High importance, Medium feasibility)
- Admin panel for user & food/recipe moderation (Medium importance, High feasibility)
- Dark mode & configurable settings (Low importance, High feasibility)
- Wearable device sync & push notifications (High importance, Low feasibility — parked for future scope)

3. Requirement Analysis

3.1 Customer Journey Map

User journey — discovering, tracking, and returning to Nutrition Assistant:

Stage	Entice	Enter	Engage	Exit	Extend
What happens	Discovers the app via search or referral	Registers/signs in and lands on the dashboard	Uses calculators, logs food & water, chats with AI, browses recipes	Reviews the daily summary; closes the app	Returns daily; checks weekly trends and adjusts goals
User goal	"Help me find a tool I can trust"	"Help me get started quickly"	"Help me stay on track confidently"	"Help me know I logged everything"	"Help me build a lasting habit"

3.2 Solution Requirement (Functional & Non-Functional)

Functional Requirements

FR No.	Functional Requirement (Epic)	Sub Requirement
FR-1	User Registration & Auth	Registration; JWT login with access/refresh tokens; forgot/reset password
FR-2	Profile Management	Demographics, goal, activity level, diet preference, allergies, photo upload
FR-3	Calculators	BMI calculator; BMR/TDEE calculator; apply calorie goal to profile
FR-4	Food & Water Tracking	Log food per meal; auto-recalculated totals; log/delete water entries
FR-5	Recipes & Meal Planning	Search/filter/paginate recipes; favorites; custom recipes; meal planner
FR-6	AI Nutrition Chatbot	Ask nutrition questions; Gemini or fallback response
FR-7	Admin Panel	Platform stats; user management; system food curation

Non-Functional Requirements

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Responsive, glassmorphic UI (React 19, Tailwind CSS v4) with dark mode
NFR-2	Security	JWT dual-token auth, bcryptjs hashing, Helmet, CORS, rate limiting, role-guard middleware
NFR-3	Reliability	Zod + Mongoose validation and centralized Express error handling
NFR-4	Performance	Vite builds, Axios interceptor layer, indexed MongoDB text search
NFR-5	Availability	Static client + persistent Node.js API backed by MongoDB Atlas
NFR-6	Scalability	Modular MVC-layered backend supports new modules without core rework

3.3 Data Flow Diagram

DFD Level-0: User → React + Vite Client → Express REST API (JWT auth → Zod validation → Controllers) → Mongoose ODM → MongoDB Atlas, with external calls to the Google Gemini API and Cloudinary.

3.4 Technology Stack

Component	Technology
User Interface	React.js 19 (Vite), Tailwind CSS v4, Framer Motion, Chart.js
Application Logic	Node.js, Express.js (MVC-layered)
Authentication & Authorization	JWT (access + refresh), bcryptjs, role-guard middleware
Database	MongoDB (Mongoose ODM), MongoDB Atlas
External API	Google Gemini API (AI Nutrition Chatbot)
File Storage	Cloudinary (local-disk fallback via Multer)
Infrastructure	Static client hosting + persistent Node.js API service

4. Project Design

4.1 Problem – Solution Fit

Users face a fragmented, low-personalization nutrition-tracking experience — separate calculators, food logs and generic advice with no shared profile. Nutrition Assistant fits this problem by giving users one platform: accurate calculators tied to their profile, effortless food/water logging with automatic totals, and an AI chatbot for instant, personalized guidance — directly addressing the fragmentation and lack of personalization that make existing tools frustrating.

4.2 Proposed Solution

Parameter	Description
Problem Statement	Fragmented nutrition tools, manual calculation, and generic (non-personalized) advice.
Idea / Solution	An integrated MERN platform combining calculators, tracking, recipes/meal planning and an AI chatbot around one user profile.
Novelty / Uniqueness	Gemini-powered AI chatbot with rule-based fallback; Mongoose pre-save hooks auto-recalculating nutrition/water totals; dynamic goal-based recommendations.
Social Impact / Satisfaction	Centralizing tracking and guidance reduces the friction that causes people to abandon diet tracking.
Business Model	Freemium — free core tracking/calculators, future premium tiers for advanced analytics and AI meal plans.
Scalability	Modular MVC backend and independently-scaling static client, ready for wearable sync, mobile app and payment integration.

4.3 Solution Architecture

Three-tier architecture: React + Vite client (presentation) → Express.js API with JWT/RBAC middleware (application) → MongoDB via Mongoose (data), with the Gemini API and Cloudinary as external services.

5. Project Planning & Scheduling

5.1 Project Planning

Product backlog (representative entries):

Sprint	Epic	User Story	Points	Priority	Team Members
Sprint-1	Registration & Login	Register and log in with JWT tokens	3	High	Naraharisetti Sai Nitchal, Sundarapalli Siva Sankar
Sprint-2	Calculators & Tracking	BMI/Calorie calculators; food & water logging	9	High	Nunna Tarun Sri Venkata Sai, SAIRAJU ADDAGARLA
Sprint-3	Recipes & AI Assistant	Recipe search/favorites, meal planner, AI chatbot	8	High	Naraharisetti Sai Nitchal, Edward Jones Devathoti
Sprint-4	Admin & Settings	Admin panel, dark mode, settings	4	Medium	SAIRAJU ADDAGARLA, Sundarapalli Siva Sankar

Sprint tracker: 4 sprints of 6 days each, 20 story points planned per sprint. Actual sprint dates and completed story points are pending real execution tracking and are not fabricated here.

6. Functional and Performance Testing

6.1 Functional / UAT Test Scenarios

Test Case ID	Scenario	Expected Result
TC-001	User registers, logs in, and views a personalized dashboard	Dashboard loads with correct calorie/water/BMI stats
TC-002	BMI Calculator computes correctly in metric & imperial	Correct BMI score and category shown in both unit systems
TC-003	Food logged updates daily nutrition totals	totalNutrition recalculates correctly via pre-save hook
TC-004	Water logged updates dashboard hydration progress	WaterLog.amount recalculates and displays correctly
TC-005	Recipe search & favorite toggle	Recipe appears in results; favorite/unfavorite works correctly

6.2 AI Chatbot Functional & Performance Testing

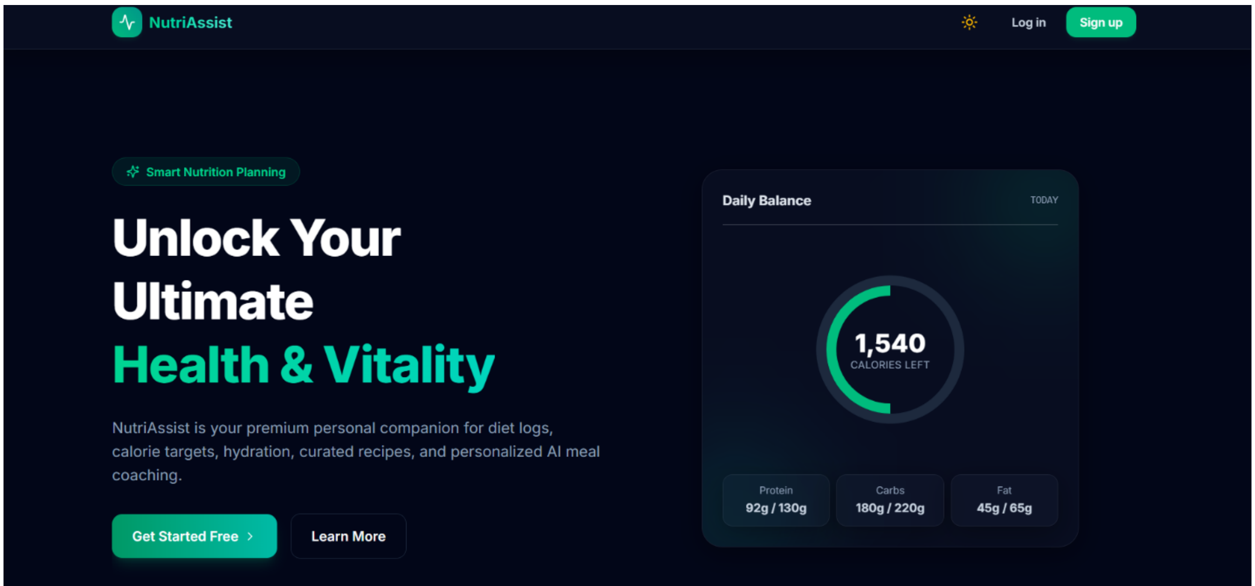
Test Case ID	Scenario	Expected Result
FT-01	Nutrition question response via /api/ai/chat	Relevant markdown response from Gemini or fallback
FT-03	Fallback behaviour without Gemini API key	Keyword-based mock response returned instead of an error
PT-01	Response time for a chatbot reply	Should be under 3–5 seconds

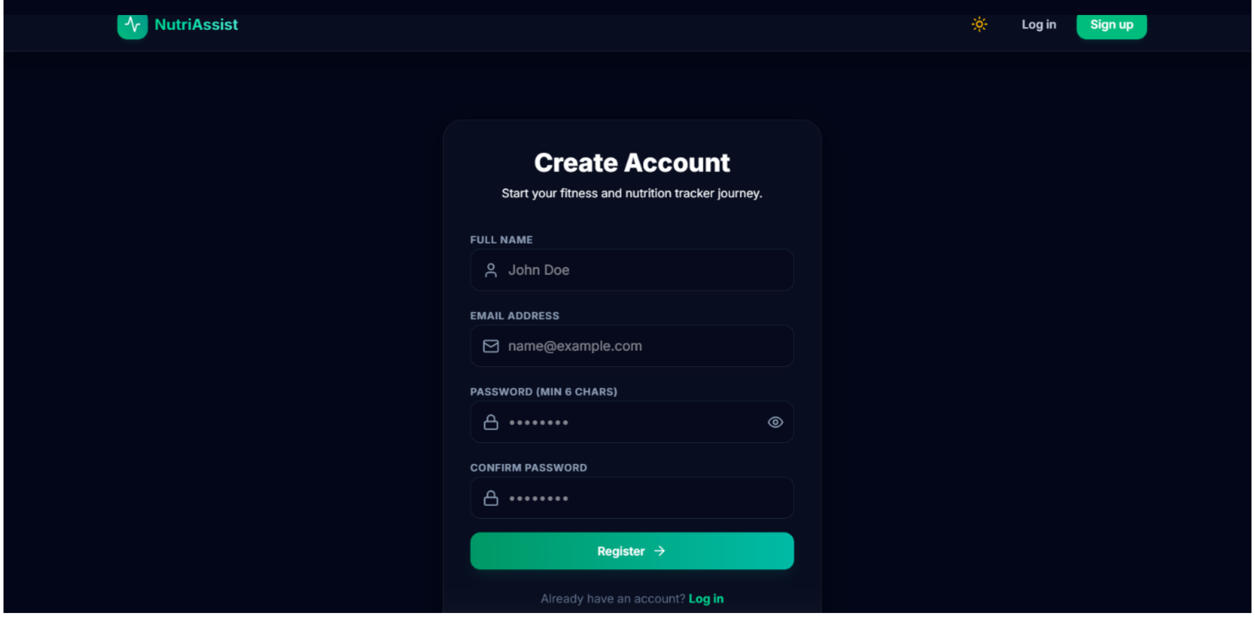
Actual/Pass-Fail results are intentionally left blank in the working templates until formal UAT execution is carried out.

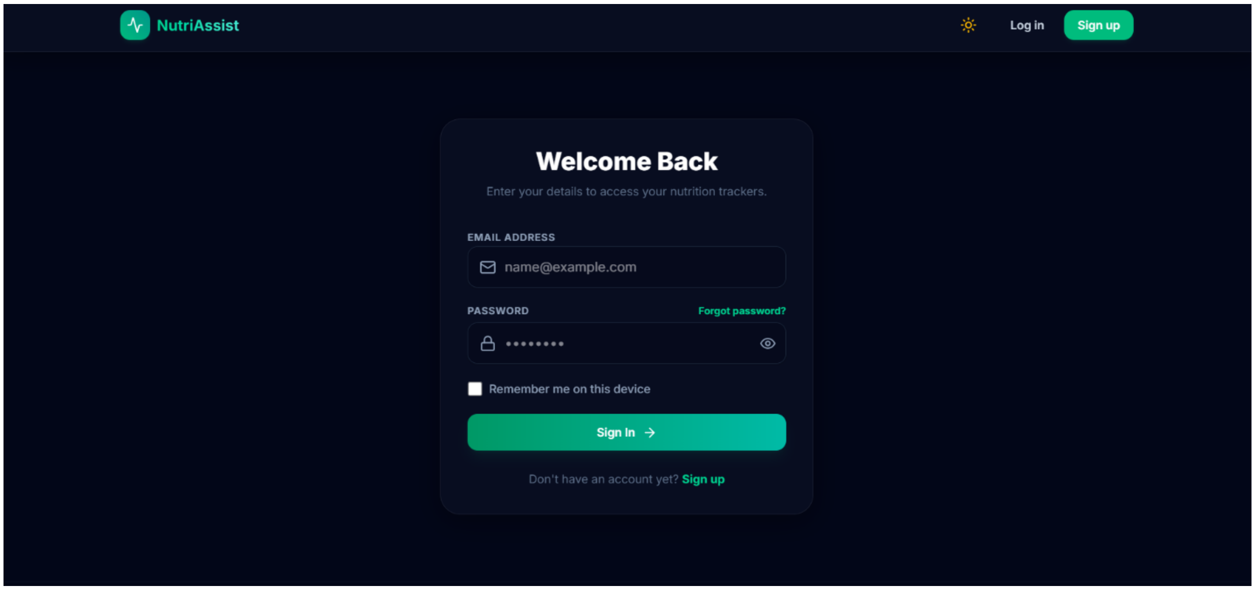
7. Results

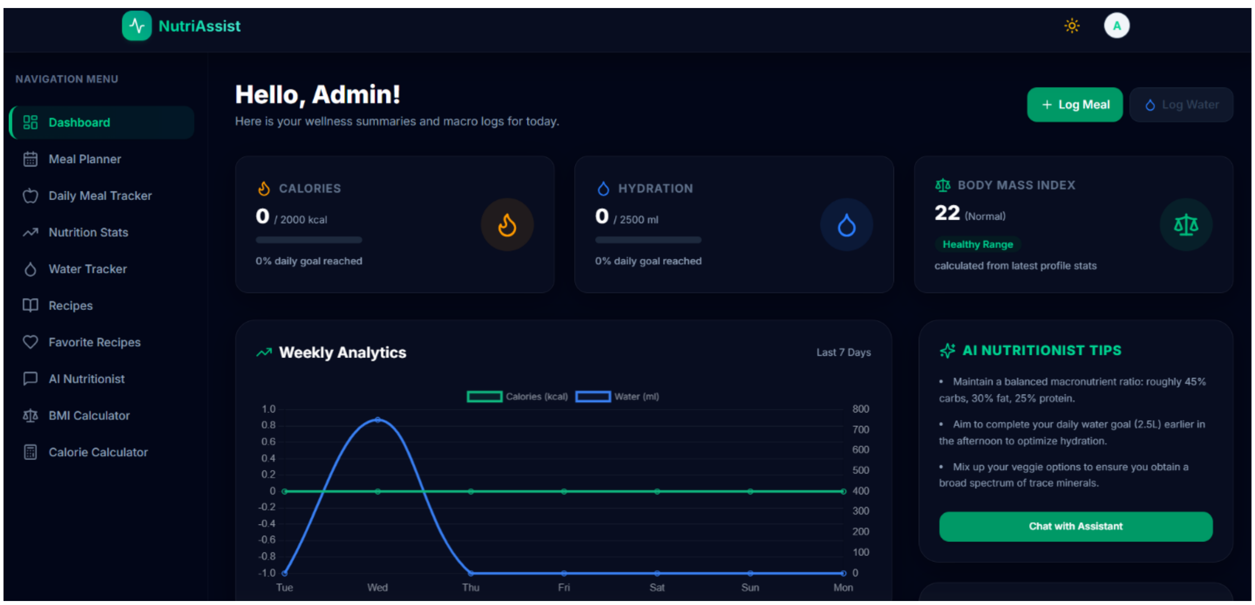
7.1 Output Screenshots

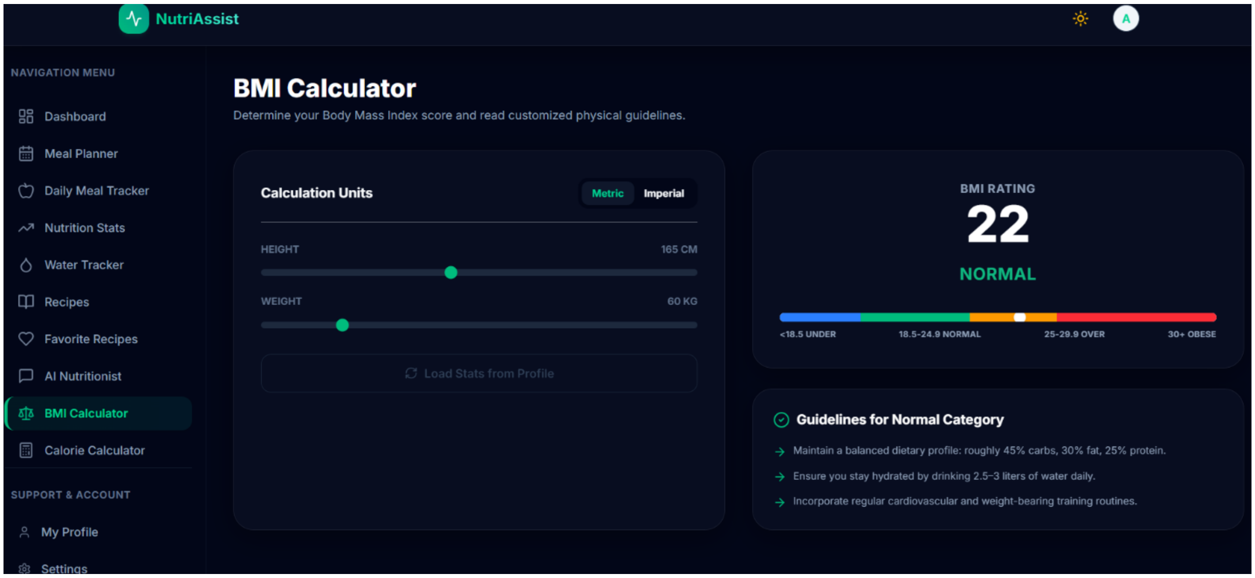
Insert screenshots of the following screens when preparing the final submission: Landing Page, Registration/Login, Dashboard, BMI Calculator, Calorie Calculator, Daily Tracker, Recipe Explorer, Meal Planner, AI Nutrition Assistant chat, Settings, Admin Dashboard, and Admin User/Food Management.






NutriAssist
Log in
Sign up
Welcome Back
Enter your details to access your nutrition trackers.
EMAIL ADDRESS
name@example.com
PASSWORD
Forgot password?
Remember me on this device
Sign In
Don't have an account yet? Sign up


NutriAssist
A
NAVIGATION MENU
Dashboard
Meal Planner
Daily Meal Tracker
Nutrition Stats
Water Tracker
Recipes
Favorite Recipes
AI Nutritionist
BMI Calculator
Calorie Calculator
Hello, Admin!
Here is your wellness summaries and macro logs for today.
Log Meal
Log Water
CALORIES
0 / 2000 kcal
0% daily goal reached
HYDRATION
0 / 2500 ml
0% daily goal reached
BODY MASS INDEX
22 (Normal)
Healthy Range
calculated from latest profile stats
Weekly Analytics
Last 7 Days
Calories (kcal)
Water (ml)
Tue
Wed
Thu
Fri
Sat
Sun
Mon
AI NUTRITIONIST TIPS
Maintain a balanced macronutrient ratio: roughly 45% carbs, 30% fat, 25% protein.
Aim to complete your daily water goal (2.5L) earlier in the afternoon to optimize hydration.
Mix up your veggie options to ensure you obtain a broad spectrum of trace minerals.
Chat with Assistant


NutriAssist
NAVIGATION MENU
Dashboard
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Daily Meal Tracker
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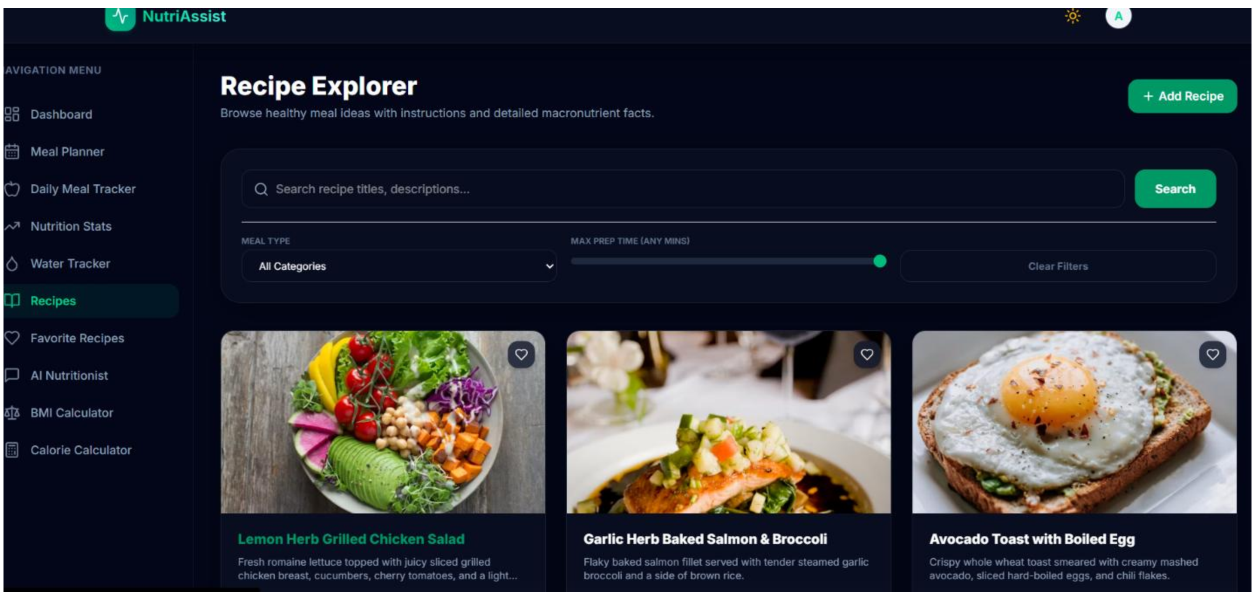
Recipes
Favorite Recipes
AI Nutritionist
BMI Calculator
Calorie Calculator
SUPPORT & ACCOUNT
My Profile
Settings
BMI Calculator
Determine your Body Mass Index score and read customized physical guidelines.
Calculation Units
Metric
Imperial
HEIGHT
165 CM
WEIGHT
60 KG
Load Stats from Profile
BMI RATING
22
NORMAL
<18.5 UNDER
18.5-24.9 NORMAL
25-29.9 OVER
30+ OBESE
Guidelines for Normal Category
Maintain a balanced dietary profile: roughly 45% carbs, 30% fat, 25% protein.
Ensure you stay hydrated by drinking 2.5-3 liters of water daily.
Incorporate regular cardiovascular and weight-bearing training routines.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AVIGATION MENU
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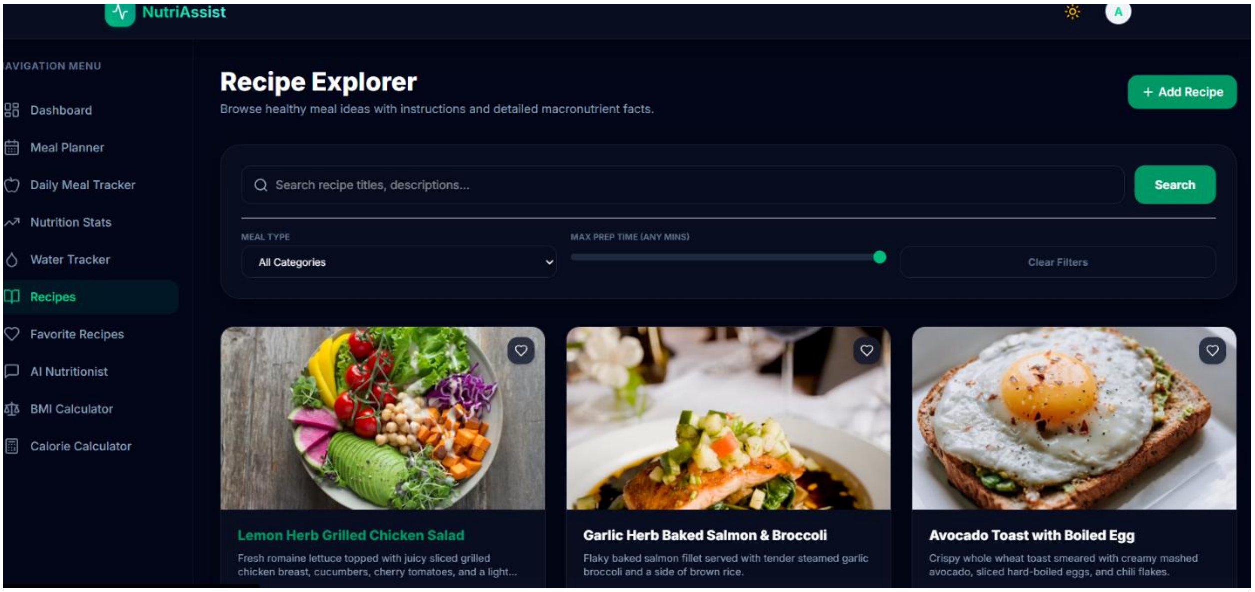
Daily Meal Tracker
Nutrition Stats
Water Tracker
Recipes
Favorite Recipes
AI Nutritionist
BMI Calculator
Calorie Calculator
Recipe Explorer
Browse healthy meal ideas with instructions and detailed macronutrient facts.
+ Add Recipe
Search recipe titles, descriptions...
Search
MEAL TYPE
All Categories
MAX PREP TIME (ANY MINS)
Clear Filters
Lemon Herb Grilled Chicken Salad
Fresh romaine lettuce topped with juicy sliced grilled chicken breast, cucumbers, cherry tomatoes, and a light...
Garlic Herb Baked Salmon & Broccoli
Flaky baked salmon fillet served with tender steamed garlic broccoli and a side of brown rice.
Avocado Toast with Boiled Egg
Crispy whole wheat toast smeared with creamy mashed avocado, sliced hard-boiled eggs, and chili flakes.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AI Nutritionist
BMI Calculator
Calorie Calculator
Recipe Explorer
Browse healthy meal ideas with instructions and detailed macronutrient facts.
+ Add Recipe
Search recipe titles, descriptions...
Search
MEAL TYPE
All Categories
MAX PREP TIME (ANY MINS)
Clear Filters
Lemon Herb Grilled Chicken Salad
Fresh romaine lettuce topped with juicy sliced grilled chicken breast, cucumbers, cherry tomatoes, and a light...
Garlic Herb Baked Salmon & Broccoli
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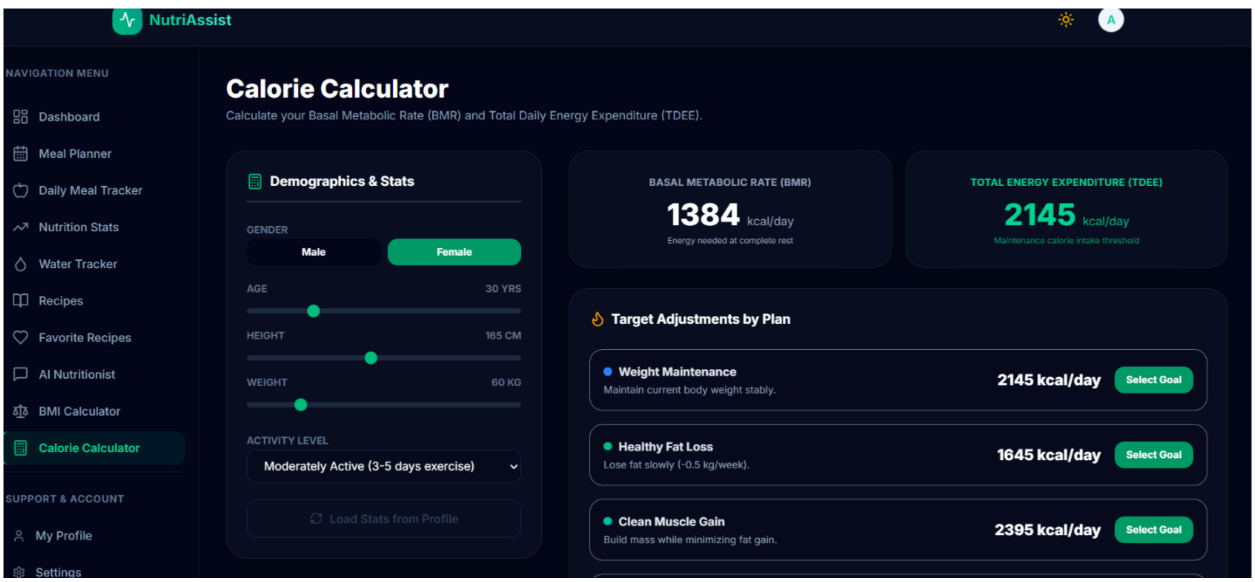
Crispy whole wheat toast smeared with creamy mashed avocado, sliced hard-boiled eggs, and chili flakes.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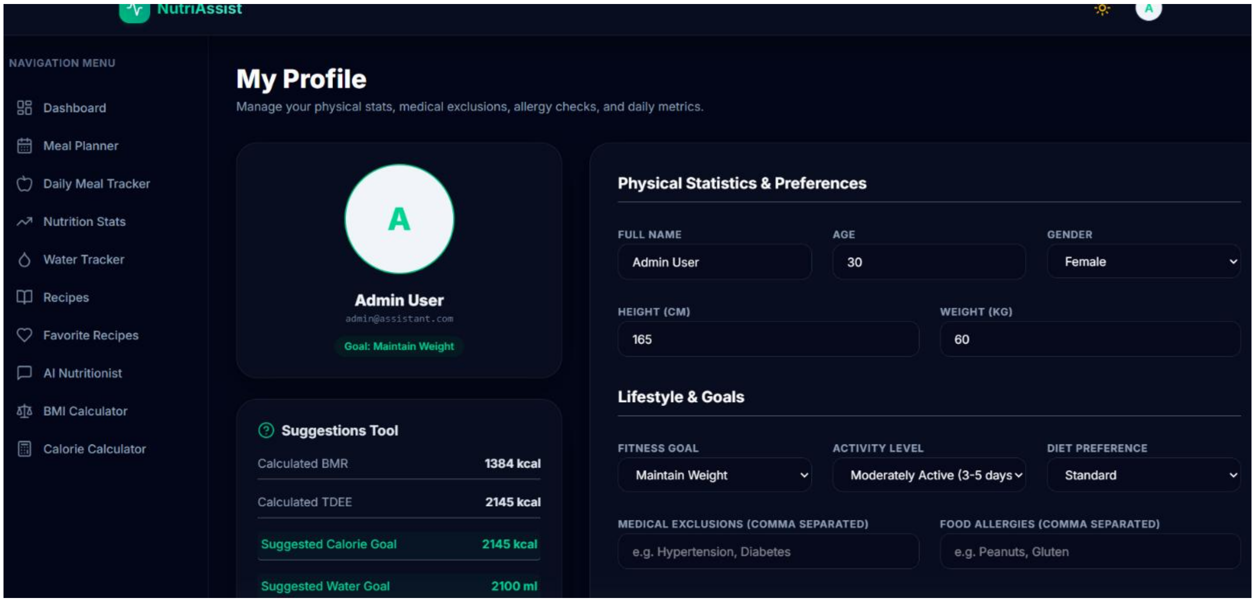
Recipes
Favorite Recipes
AI Nutritionist
BMI Calculator
Calorie Calculator
SUPPORT & ACCOUNT
My Profile
Settings
Calorie Calculator
Calculate your Basal Metabolic Rate (BMR) and Total Daily Energy Expenditure (TDEE).
Demographics & Stats
GENDER
Male
Female
AGE
30 YRS
HEIGHT
165 CM
WEIGHT
60 KG
ACTIVITY LEVEL
Moderately Active (3-5 days exercise)
Load Stats from Profile
BASAL METABOLIC RATE (BMR)
1384 kcal/day
Energy needed at complete rest
TOTAL ENERGY EXPENDITURE (TDEE)
2145 kcal/day
Maintenance calorie intake threshold
Target Adjustments by Plan
Weight Maintenance
Maintain current body weight stably.
2145 kcal/day
Select Goal
Healthy Fat Loss
Lose fat slowly (-0.5 kg/week).
1645 kcal/day
Select Goal
Clean Muscle Gain
Build mass while minimizing fat gain.
2395 kcal/day
Select Goal



8. Advantages & Disadvantages

Advantages

- Single platform combining calculators, tracking, recipes and AI guidance, replacing scattered manual tools
- Automatic nutrition/water total recalculation removes manual math from daily tracking
- Gemini-powered AI chatbot with a working fallback keeps guidance available even without an API key
- JWT dual-token auth with role-based access keeps user and admin capabilities securely separated
- Modular MVC architecture makes the platform straightforward to scale and extend

Disadvantages / Current Limitations

- No online payment or premium subscription tier is integrated yet
- Password reset tokens are stored in-memory and are lost on server restart
- The configurable reminder interval does not yet trigger push/email notifications
- Admin food management supports one-by-one actions rather than bulk operations
- Local-disk image fallback is not persistent across some deployment platforms

9. Conclusion

Nutrition Assistant is a complete MERN-stack web application that brings BMI/calorie calculation, food and water tracking, recipe discovery, meal planning, and AI-powered nutrition guidance together into a single, cohesive platform. By combining secure authentication, automatic nutrition aggregation, responsive design and an integrated AI chatbot, Nutrition Assistant lays a strong foundation for evolving into a comprehensive, AI-assisted personal wellness ecosystem.

10. Future Scope

- AI-personalized weekly meal plans generated from logged history
- Wearable device integration for automatic activity and calorie-burn tracking
- Push and email notification system for reminders and goal summaries

- Native mobile application (iOS & Android)
- Persistent, database-backed password reset token storage
- Premium subscription tier with advanced analytics

11. Appendix

Source Code

Server: server/ (Node.js + Express, models / controllers / routes / middleware / validators / services / utils)

Client: client/ (React + Vite, pages / contexts / layouts / components / services)

Dataset Link

Not applicable — Nutrition Assistant uses live application data (foods, recipes, logs) seeded automatically into MongoDB via npm run seed, rather than a static training dataset.

GitHub & Project Demo Link

GitHub repository: [add your repository URL here]

Deployed application: [add your deployed application URL here]

Demo accounts — Admin: admin@assistant.com / admin123 · User: user@assistant.com / user123